A Project Report on

E-commerce Website for Janardanswami Yogabhyasi Mandal

Submitted To
Department of Information Technology Government Polytechnic, Nagpur in the partial fulfilment of the requirements for the award of the

Diploma in Engineering
(Information Technology)

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have completed the project work under my guidance and supervision and I have verified the work for its originality in documentation, problem statement, implementation and results presented in the dissertation. Any reproduction of other necessary work is with prior permission and has given due ownership and is included in the references.

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Date: 5th May 2022

Respected Madam,

I am founder of Auriga AI Solutions Pvt ltd, which specializes as an AI Solutions company. Earlier I approached students from the Government Polytechnic (IT department) to work on a mini project with me which focuses on developing a conversational AI Model for restaurants. The following students are being guided by me, Janhavi N. Itankar, Aayushi A. Sonkusare, Uzma Jabeen Syed Raza Ali and Varad V. Nimbarte

Thank you.

Saumitra Mahajan

Founder,

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Ku.Uzma Jabeen Syed Raza Ali
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Abstract

Janardanswami Yogabhyasi Mandal is a non-governmental organisation working on providing free Yoga classes for people of all age groups. They also provide free diagnosis for diseases and recommend *Yogasana* for the same.

This organisation also provides books for sale based on a variety of topics such as Yoga postures, Yoga for specific diseases, monthly and fortnight magazines, etc. This books are currently sold only via their organisation's centre at Ram Nagar and through postal services. This current approach is non-scalable and it is limited to only people from Nagpur mostly, as there is no proper way for people outside Nagpur to buy the books.

For solving this problem we decided to use the approach of Designing, Developing and Deploying an E-commerce website where it would be possible for people in remote areas to buy the books with the minimum requirement of internet, and a working web browser. As this would be an ideal approach and scalable according to the requirements of the organisation.

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Notation and Abbreviations

Abbreviation	Description
IDE	Integrated Development Environment
Vue	Vue js
Git	Version control system
VS	Visual studio code
Figma	UI Designing software
Tailwind	Utility-based CSS framework
CRUD	Create, Read, Update, Delete
SRS	Software Requirement Specification

Table 1: Abbreviation table

Chapter 1

Introduction

A website can be likened to a makeover for a company or organisation. It helps to raise more funds while also attracting more people's attention. A well-designed website has the ability to entice potential clients in a variety of ways.

Excellent information, simple navigation, and the greatest services are just a few examples. The main purpose of e-commerce is to reach as many customers as possible at the correct time. It's time to boost your company's sales and profits. E-commerce involves purchasing and selling things as well as sending money or data over the internet.

1.1 What is Janardanswami Yogabhyasi Mandal?

Organizations like Janardan Swami Yogabhyasi Mandal are a rarity in today's utilitarian society. It is a one-of-a-kind organisation that promotes 'Yog' as a way of life or culture in society for the contentment, health, and happiness of all souls by providing free Yog training to anybody, regardless of caste, creed, or religion. It is the organization's main goal and mission, as well as the phrase encoded in its logo,

which is the message of the organization's founder, parampujya yog murti Janardhan Swami, The Divine Saint, Guru, The Source of Inspiration, and the organization's principal guiding force. Swamiji was a great missionary with vision. He rightly recognized that man is bound to suffer from the acute stresses of high-speed, high-tech modern society which is full of materialistic excesses, avarice, indiscipline, inherent contradictions, bad habits which would cause several psychosomatic disorders & ailment. However he was confident that he had single but the most effective solution for all these. Swamiji was always on the move for the cause of yog. He was full of compassion for fellow beings. He always rushed to the rescue of the

sufferer of chronic ailments, who had tried all medicines, in vain. He used to take distant walks to reach to the sufferer right at the appointed time to teach some yogic asanas or kriyas. Usually the blessed person used to get substantial relief and get into a convert to yogic way of life and devout admirer of Swamiji, Numerous experiences of such type are vouched by many beneficiaries even now.

1.2. About Our Website

Janardanswami Yogabhyasi Mandal is a non-governmental organisation working on providing free Yoga classes for people of all age groups. They believe that yoga can cure all the problems in the world. They also provide free diagnosis for diseases and recommend *Yogasana* for the same.

The Janardanswami Yogabhyasi Mandal organisation also provides books for sale based on a variety of topics such as Yoga postures, Yoga for specific diseases, monthly and fortnight magazines, etc for betterment of the humans and help them cure their physical and mental issues by expert solutions. These books are currently sold only via there organisation's centre at Ram Nagar and through postal services. This current approach is non-scalable and it is limited to only people from Nagpur mostly, as there is no proper way for people outside Nagpur to buy the books as the people mostly need to come physically and choose the books as there are no means for giving a short overview of the book or taking a overlook of the book cover.

For solving this problem we decided to use the approach of Designing, Developing and Deploying an E-commerce for website for selling the books provided by the mandal. This website will make possible for people in remote areas to buy the books with the minimum requirement of internet, and a working web browser. As this would be an ideal approach and scalable according to the requirements of the organisation.

People can search books based on the filters as price or category. View the details of the book and description. Add the books to their cart directly or keep them as an option for future purpose in the Wishlist. As the order will reach the customer by postal service the order id of the post services will be given to the customer so they can track their package. The website will contain all the books that are sold at the Ram nagar centre by the mandal. The website provides all the functionalities provided by the most e-commerce sites but exclusively for the Janardanswami Yogabhyasi Mandal.

1.3 Existing Book Sites:

There are many book selling sites present on the internet like:

1. Amazon: the amazon has created a special system dedicated to books. All types of books are available on this site for the users.

2. Flipkart : Along with other products flipkart has been selling books for years and is been providing a good services to the user.

3. Snapdeal: Altough snapdeal is not much dedicated to books, we can see a lot of books of various genre. There are many more sites for books.

The Janardanswami Yogabhyasi Mandal decided not to use these platforms for selling their books as they needed a totally dedicated site for their books so that they can provide the best books to their customers based on their need.

Chapter 2

Background And Literature Survey

2.1 Feasibility Report:

A detailed investigation and analysis conducted to determine the financial, economic, technical, or other advisability of a proposed project. Part of the systems development life cycle which aims to determine whether it is sensible to develop some system. The most popular model of feasibility study is "TELOS", standing for Technical, Economic, Legal, Operational, Schedule. Technical Feasibility: The software required to develop the system as well as to run it are easily accessible and free of cost and hence it is technically feasible. Economic Feasibility: The system is simple, user-friendly set up and personalization, cost-efficient/free, easily accessible, and hence it is economically feasible.

Legal Feasibility: There is no conflict between the proposed system and legal requirements. Operational.

Feasibility: The current work practices and procedures adequate to support the new system. the system is simple to use, easily accessible and free of cost and hence it is operationally feasible. Schedule Feasibility: The system is developed within time limit.

2.2 Innovativeness and Usefulness:

- ✓ The proposed system will provide people a platform to buy books of their choice and benefit from their home itself.
- ✓ The system will provide a centre stage for the Yogabhyasi mandal to publish their new launches of books or fortnight magazines.
- ✓ The site will merge with the main site of the mandal so it could be easily accessible to the people who wish to order books from the site.
- ✓ The books on the site would be totally related to the physical and mental health of people that would aware people to become health conscious

2.3 Market Potential and Competitive advantages:

- ✓ Ease of access to old/new books published by the mandal
- ✓ Books can be purchased from remote locations
- ✓ Larger group of people can visit the site and show their interest in books
- ✓ Simple, user-friendly
- ✓ cost efficient/free
- ✓ Easily accessible
- ✓ Scalability

Chapter 3

Software Requirement Specification

3.1 Introduction

Organizations like Janardan Swami Yogabhyasi Mandal are a rarity in today's utilitarian society. It is a one-of-a-kind organisation that promotes 'Yog' as a way of life or culture in society for the contentment, health, and happiness of all souls by providing free Yog training to anybody, regardless of caste, creed, or religion. It is the organization's main goal and mission, as well as the phrase encoded in its logo, which is the message of the organization's founder, parampujya yog murti Janardhan Swami, The Divine Saint, Guru, The Source of Inspiration, and the organization's principal guiding force. his organisation also provides books for sale based on a variety of topics such as Yoga postures, Yoga for specific diseases, monthly and fortnight magazines, etc.

Purpose

These books are currently sold only via there organisation's centre at Ram Nagar and through postal services. This current approach is non-scalable and it is limited to only people from Nagpur mostly, as there is no proper way for people outside Nagpur to buy the books. For solving this problem, we decided to use the approach of Designing, Developing and Deploying an E-commerce website where it would be possible for people in remote areas to buy the books with the minimum requirement of internet, and a working web browser. As this would be an ideal approach and scalable according to the requirements of the organisation.

3.1.1 Definitions, Acronyms, and Abbreviations

A website can be likened to a makeover for a company or organisation. It helps to raise more funds while also attracting more people's attention. A well-designed website has the ability to entice potential clients in a variety of ways. Excellent information, simple navigation, and the greatest services are just a few examples. The main purpose of e-commerce is to reach as many customers as possible at the correct time. E-commerce involves purchasing and selling things as well as sending money or data over the internet.

Acronyms

- **NoSQL:** Not SQL, queries involving collections and documents and not the legacy SQL Syntax
- **Vue.js:** A progressive framework for building user interfaces.
- **TailwindCSS:** A utility-first CSS framework for rapidly building custom user interfaces.
- **API:** Application Programming Interface
- **CRUD:** Create Update Delete operation

3.1.3 Document Convention

The document has used underlined words to highlight the user requirements. It also uses capital lettered words to highlight key words. The document has tried to maintain a priority of requirements. The priority has been determined by the judgment of the author and may subject to change. Priority of higher-level requirements is inherited by detailed requirements. The document has used short forms for some commonly abbreviated terms.

3.1.4 Intended Audience and Reading Suggestions

The document is intended to be a guide for developers, users and testers. The goal of this document is to identify the requirements of website e-commerce. A developer may directly want to know the requirements by reading the section of USER REQUIREMENTS and skipping other sections if required.

3.1.5 Product Scope

❖ **Purpose**

The objective of this software is to completely automate the following processes.

- The main objective of the project is to establish a network among the people residing in rural or remote areas.
- This system provides users to register themselves in profile like name, email and address.
- This system provides users the ability to buy books they wish while sitting at their home.
- This system provides user to join the organization according to their scenario.
- This system provides the user to maintain their Wishlist and user can update their Wishlist.
- This system provides user to keep a tab on bestselling books to join the community of readers.

❖ **Benefits**

- One of the many advantages is that you can connect with the organization, all members of e-commerce website. Simply type in the name of the book and you get the details of the book.
- As if that wasn't enough, our site is just an easy way to relax. You can order on our web page and then be free of any hassle of picking up the parcel.
- Staying in touch with contacts is a drag, so without ant useless connection, buy books and be free of any other disturbance.

❖ **Objectives**

- A goal or objective is a projected computation of affairs that a person or a system plans or intends to achieve—a personal or organizational desired end-point in some sort of assumed development. Many people endeavour to reach goals within a finite time by setting deadlines.

- It is roughly similar to purpose of aim, the anticipated result which guides reaction, or an end, which is an object, either a physical object or an abstract object, that has intrinsic value.

3.2 Overall Description

3.2.1 Product Perspective

E-commerce websites make people more managed and help them have their hobbies/passions fulfilled without going out of the way. E-commerce websites are places where people can maintain and nurse their existing income and create new (online) opportunities. E-commerce websites are places where people learn the crucial importance of being able to control their extravagance which they can benefit from in their future professional life.

3.2.2 Product Functions

- ❖ Maintenance of detail of various users.
- ❖ Online circulate the information.
- ❖ Status of all books and genre details easily available.
- ❖ Buying and updating of books for remote areas.

3.2.3 Operating Environment

Processor	**Pentium 500 MHz**
RAM	**128 MB or More**
Hard Drive	**20 MB Disk Drive Space (Available)**
Internet Connection	**High speed internet connection**

Table 1: OS Environment

3.2.3 Design and Implementation Constraints

The project operates under several design and implementation constraints. Some of these are as outlined below:

- **Hardware and Software constraints:**

  Since the project has been developed entirely using TailwindCSS and VueJS methodology, it is largely unresponsive. The project can be run on any platform. However, it still needs to be at desktop site to completely show the alignments.

- **End user constraints:**

  As mentioned above the major constraints for the end user is having a computer display. Also, he should be familiar with the operation with the system to a certain extent, although a help document will be provided to oriented users.

3.2.4 Assumptions and Dependencies

The proposed solution will be designed to work in an enterprise environment. The target environment may consist of wired and wireless links inside the network. The solution has to be self-sufficient and free from any unfamiliar dependencies.

3.3. System Features

A website has divided its working in different webpages. Each module consists of different working for proper functioning of the website. All modules are integrated together to make robust website. These modules provide various features of system.

- Login/Register module
- Homepage

- Search Option
- Add book to Cart or Wishlist
- Check Order Details
- Check Order History
- Book Details

3.4. External Interface Requirements

3.4.1 User Interface

A graphical User interface is available providing following functionalities:

- Home page –all books will show here
- Login/Register- the user can login or register from this page
- User profile –user can edit its profile
- Search –user can search for books
- Wishlist/Cart- the user can see the books added in their Wishlist or Cart
- Order Details/Order History- user can see the books they've ordered or did order before in these pages
- Book View- Details of the book will show here

3.4.2 Hardware Interfaces

The solution extensive use of several hardware devices. These devices include;

- Internet modem
- LAN
- Switches

3.4.3 Software Interface

Operating system –

With minor allowances for the peculiarities of particular operating systems, the application can be modified to act largely platform independent. The

application has however been developed using Windows 11 and will naturally work without changes on the operating systems.

3.4.4 Communications Interface

As the website is e-commerce web site it requires high speed internet modem for the use of this payment procedure. For the suitable use there must be a correct internet connection among the users.

3.5. Other Non-functional Requirements

3.5.1 Performance Requirements

The solution has to exhibit very stringent performance requirements**.** The system has very perfect performance. It can help the organization to show all the books they have when it can be sold out to the members. To the user it makes it easier to plan ahead and make a schedule about the book arriving.

3.5.2 Safety Requirements

- **Security**

  - Some security measures are provided to the website account holders such as account holder must give his/her account id and password to login. Other than that security to user's personnel details and photos.

3.5.3 Software Quality Attributes

- **Availability**

This website in useful in making relationship among the organization, and the members so that the community stays put.

In such instances, there strength arise among others from

- A unifying purpose and clear coordination structure

- Flexibility
- Dynamism and creativity
- Quality and packaging of books

3.6. Other Requirements

Appendix A: Glossary

LAN=Local Area Network.

GUI=Graphical User Interface

Chapter 4

System Design

4.1 Use Case Diagram

Use-case diagram is used to define functions of the system and to capture and analyse the functional requirements of an application system. A use case model is a collection of use cases which specify the behaviour of a system.

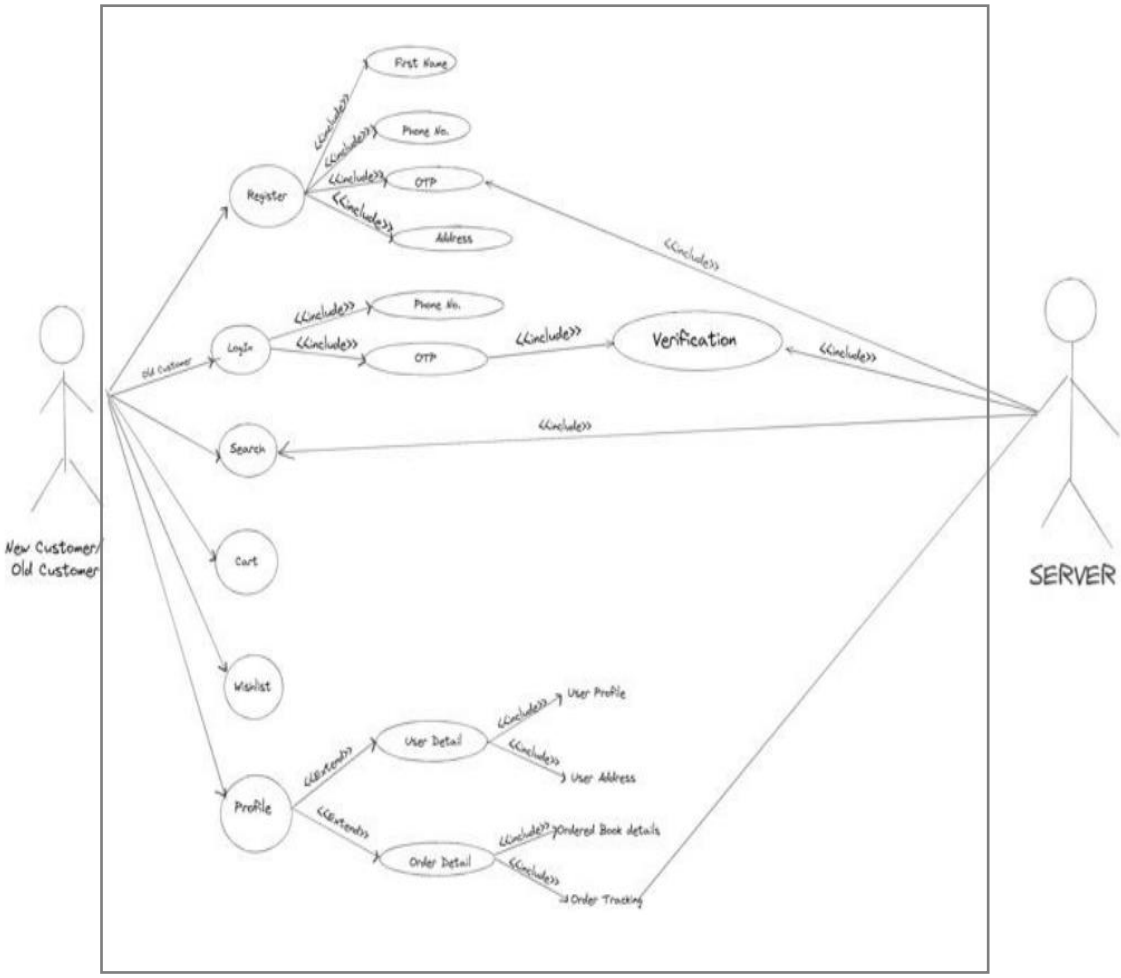

Fig 4.1: Use Case Diagram

4.2 Database Diagram

A database model shows the logical structure of a database, including the relationships and constraints that determine how data can be stored and accessed. Individual database models are designed based on the rules and concepts 318 of whichever broader data model the designers adopt. Most data models can be represented by an accompanying database diagram.

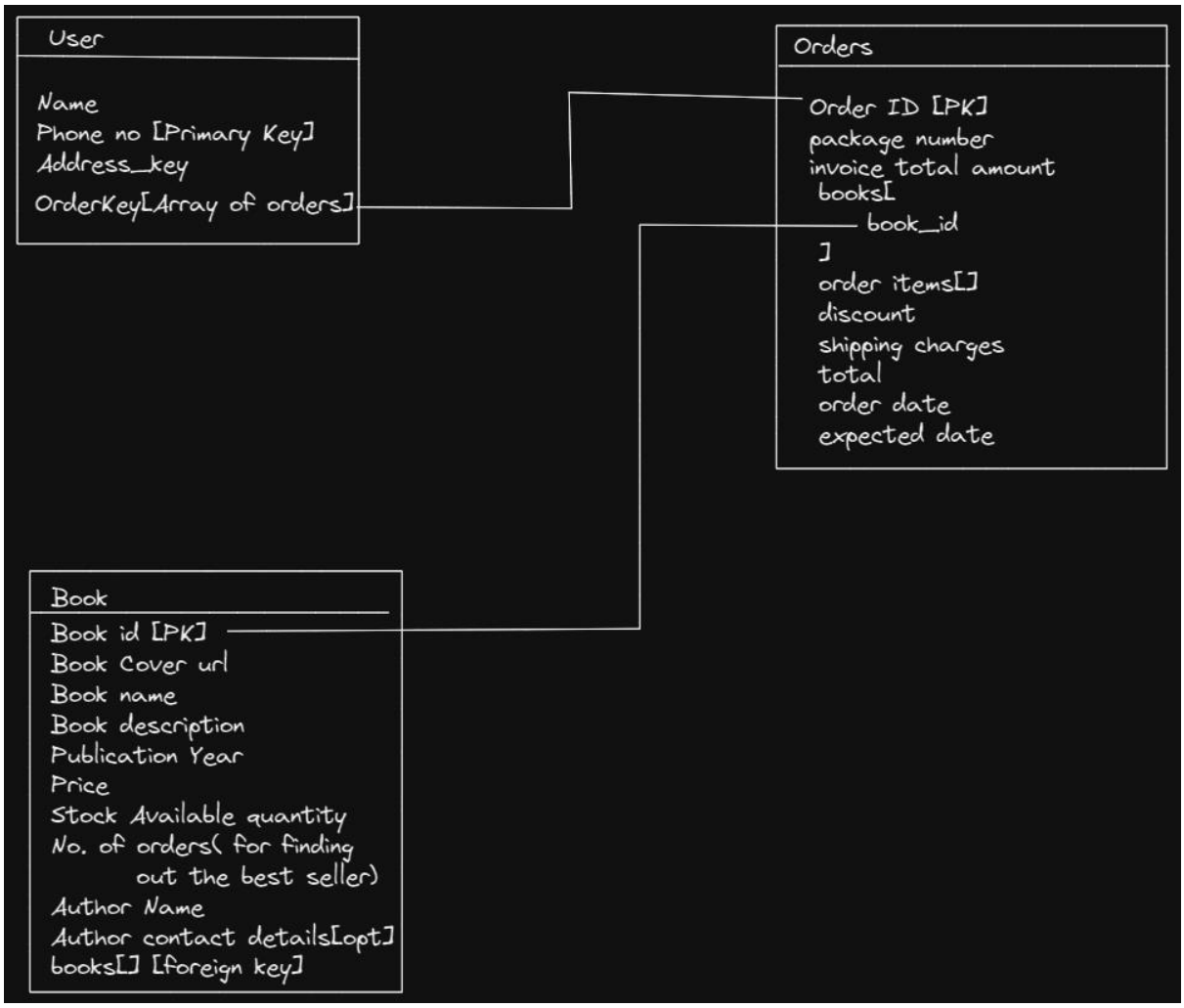

Fig 4.2: Database Diagram

4.3 Flow Diagram

A data flow diagram is a way of representing a flow of data through a process or a system, usually an information system. The Dataflow diagram alse provides information about the outputs and inputs of each entity and the process itself. A data flow diagram has a no control flow, there are no decisions rules and no loops. Specific operations based on the data can be represented by a flowchart.

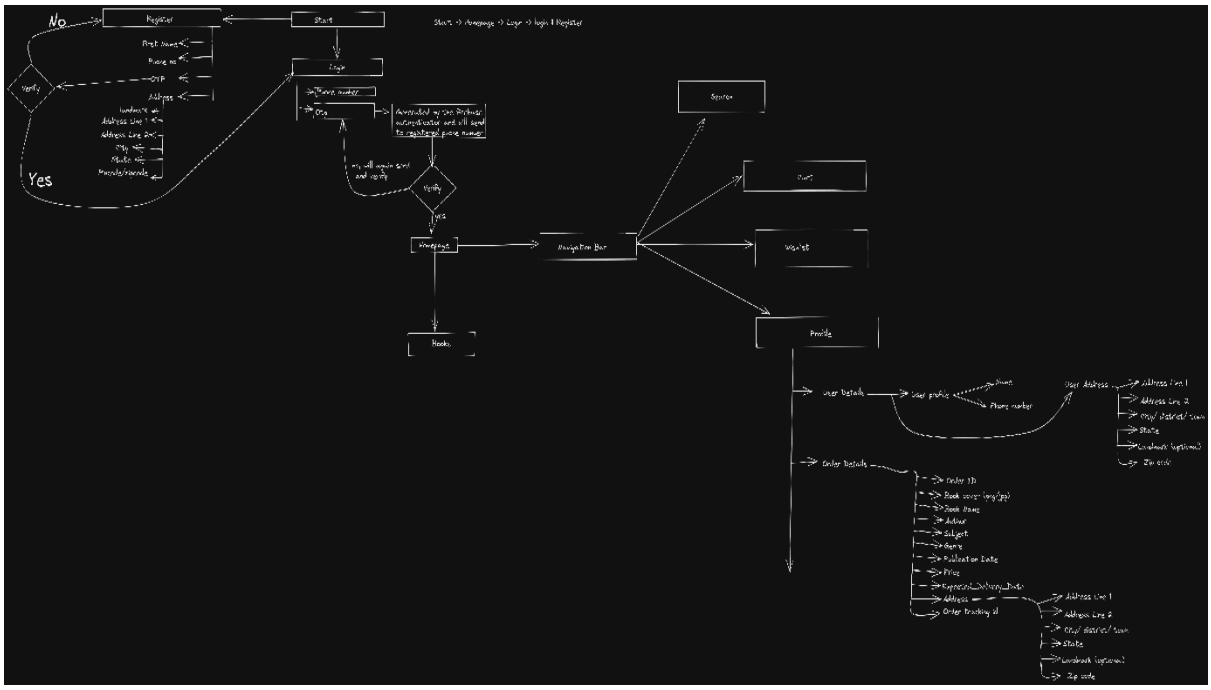

Fig 4.3: Flow Diagram

4.4 Spiral Model

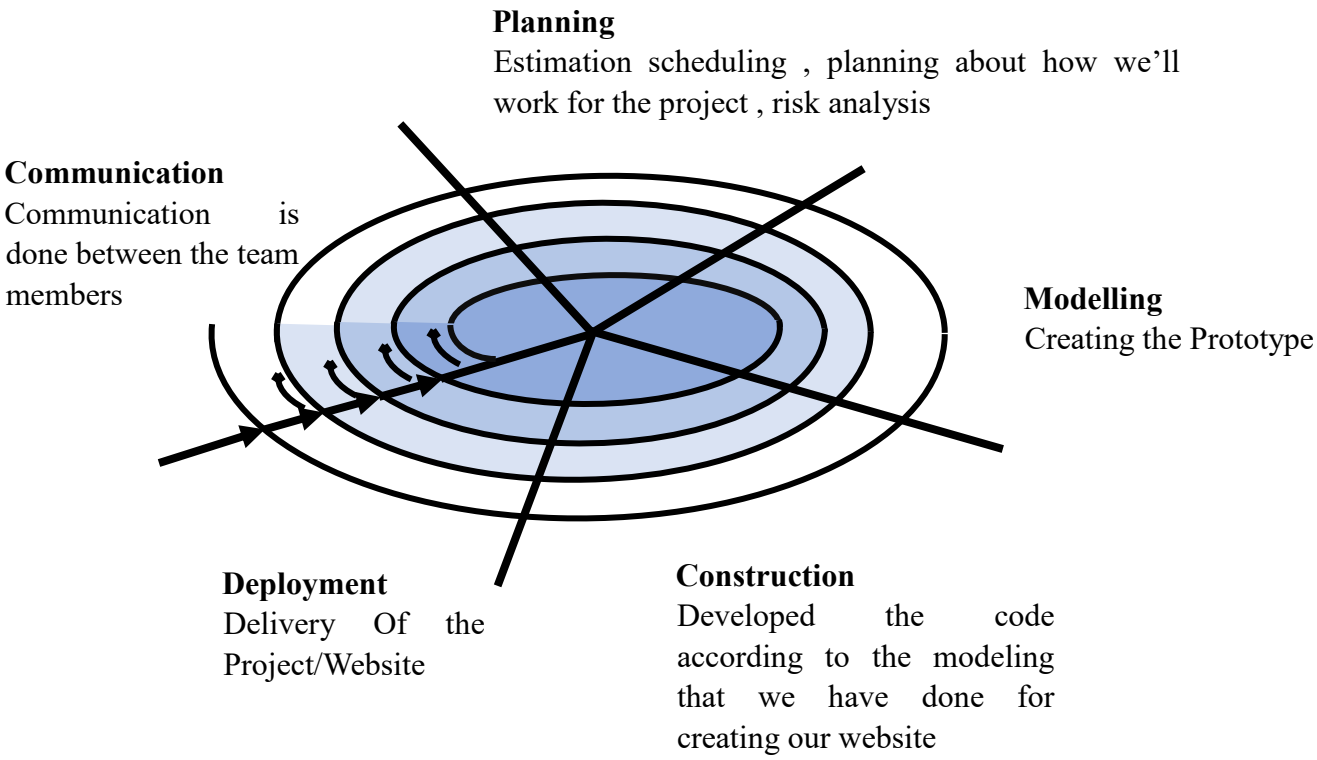

Fig 4.4: Spiral Model Diagram

1) Communication Phase

In communication phase, we communicate with our team member and client, what he actually Requires, between communication we come up with website development on an Ecommerce Website for Books which we named as Yoga Abhyas Mandal.

We discuss the functionality of website, client Requirements in website developing, GUI.

With team members we communicate with our client also, how we are going to Develop website which attributes are used in it.

- Discuss the time for completion of project which is in 3 weeks that is from 11 March to 26 March .
- Selection of modules and functionality that we are going to present in our GUI.
- Communicating with client, we discuss which kinds of book he wants to display on website specially.
- We communicate to understand requirements of client and all the functional, non-functional requirement also.

2) Planning Phase

In planning phase, first we divide the team member and allot work to them. After dividing the tasks, we plan, by using spiral model we are going to develop website. Each phase is one after another i.e. in a linear Fashion.

As per task distribution, we distribute coding among all the team members that is of front end and back end.

- We Discuss Which Resources Are Required for completing the project.
- Estimation of time and cost.

• Time Estimation

- We will make Gantt chart for scheduling of task as per the time. Refer Fig.1.
- Divide the task as per the time, first day is given to communication, planning. second day is given to modelling of GUI, third day to testing and fourth of deployment.
- Discuss chances of Risk came between the project and how to deal with it, if it became in real.

Tasks	Duration (in days)
Communication	From 11 March to 26 March
Planning	From 26 March to 9 April
Modeling	From 16 April to 30 April
Testing	From 7 May to 14 May
Deployment	On 25th May

Table 3: Time Estimation

3) Modeling Phase

Our team member Created a simple prototype through which we can better understand that how our actual website looks like. The model contains of the design of all the pages that we include in our website. We communicate with the client that which type of color combination does he wants to add through all over the website, what kind of options and features do they required. After discussing all these things with the client we come to know how we have to create our prototype/model based on the clients requirement.

Modeling Phase include

1) Analyzing :-

- We analyze the need of Module.
- We analyze working of each module.
- We made a complete analysis on estimation of time and cost and expected to deliver our final project on time.

2) Design: -

- Modules were decided and distributed among the members.
- Time was allotted accordingly.
- Complete designing of our GUI was made in this section.

4) Construction

- Our team leader Allot each members from a team to construct the code GUI of a website.
- Each and every members have allotted with each page.
- All the pages are successfully created by the team members then all those are handover to the team leader then for the further process.
- With the help of visual studio app we create the dynamic localhost website.

5) Deployment

In this phase, we finally deliver our project to our client

- Feedback on the GUI will be given by our client
- Depending on Feedback changes could be made

Chapter 5

Technical Specifications

1. Vue.js

Vue is a JavaScript framework for building user interfaces. It builds on top of standard HTML, CSS and JavaScript, and provides a declarative and component-based programming model that helps you efficiently develop user interfaces, be it simple or complex.

Vue is a framework and ecosystem that covers most of the common features needed in frontend development. But the web is extremely diverse - the things we build on the web may vary drastically in form and scale. With that in mind, Vue is designed to be flexible and incrementally adoptable. Depending on your use case,

Vue can be used in different ways:

- Enhancing static HTML without a build step
- Embedding as Web Components on any page
- Single-Page Application (SPA)
- Fullstack / Server-Side-Rendering (SSR)
- Jamstack / Static-Site-Generation (SSG)
- Targeting desktop, mobile, WebGL or even the terminal

2. Tailwind CSS

Tailwind CSS is a "utility-first" CSS framework that provides a deep catalogue of CSS classes and tools that lets you easily get started styling your website or application. The underlying goal is that as you're building your project, you don't need to deal with cascading styles and worrying about how to override that 10-selector pileup that's been haunting your app for the last 2 years.

Tailwind's solution is to provide a wide variety of CSS classes that each have their own focused use. Instead of a class called .btn that is created with a bunch of CSS attributes directly, in Tailwind, you would either apply a bunch of classes like bg-blue-500 py-2 px-4 rounded to the button element or build a .btn class by applying those utility class to that selector.

3. NOSQL

NoSQL databases (aka "not only SQL") are non-tabular databases and store data differently than relational tables. NoSQL databases come in a variety of types based on their data model. The main types are document, key-value, wide-column, and graph. They provide flexible schemas and scale easily with large amounts of data and high user loads.

As storage costs rapidly decreased, the amount of data that applications needed to store and query increased. This data came in all shapes and sizes — structured, semi-structured, and polymorphic — and defining the schema in advance became nearly impossible. NoSQL databases allow developers to store huge amounts of unstructured data, giving them a lot of flexibility.

Each NoSQL database has its own unique features. At a high level, many NoSQL databases have the following features:

- Flexible schemas
- Horizontal scaling
- Fast queries due to the data model
- Ease of use for developers

4. Git

Git is a free and open-source distributed version control system designed to handle everything from small to very large projects with speed and efficiency.

Git is easy to learn and has a tiny footprint with lightning-fast performance. It outclasses SCM tools like Subversion, CVS, Perforce, and ClearCase with features like cheap local branching, convenient staging areas, and multiple workflows.

5. GitHub

At a high level, GitHub is a website and cloud-based service that helps developers store and manage their code, as well as track and control changes to their code. To understand exactly what GitHub is, you need to know two connected principles:

- Version control
- Git

GitHub is a for-profit company that offers a cloud-based Git repository hosting service. Essentially, it makes it a lot easier for individuals and teams to use Git for version control and collaboration.

GitHub's interface is user-friendly enough so even novice coders can take advantage of Git. Without GitHub, using Git generally requires a bit more technically savvy and use of the command line.

GitHub is so user-friendly, though, that some people even use GitHub to manage other types of projects – like writing books. Additionally, anyone can sign up and host a public code repository for free, which makes GitHub especially popular with open-source projects.

As a company, GitHub makes money by selling hosted private code repositories, as well as other business-focused plans that make it easier for organizations to manage team members and security.

6. CRUD

Within computer programming, the acronym CRUD stands for create, read, update and delete. These are the four basic functions of persistent storage. Also, each letter in the acronym can refer to all functions executed in relational database applications and mapped to a standard HTTP method, SQL statement or DDS operation.

It can also describe user-interface conventions that allow viewing, searching and modifying information through computer-based forms and reports. In essence, entities are read, created, updated and deleted. Those same entities can be modified by taking the data from a service and changing the setting properties before sending the data back to the service for an update. Plus, CRUD is data-oriented and the standardized use of HTTP action verbs.

7. FirebaseFirestore

Firestore is a NoSQL document database built for automatic scaling, high performance, and ease of application development. While the Firestore interface has many of the same features as traditional databases, as a NoSQL database it differs from them in the way it describes relationships between data objects

8. FirebaseAuth

Firebase Authentication provides backend services, easy-to-use SDKs, and ready-made UI libraries to authenticate users to your app. It supports authentication using passwords, phone numbers, popular federated identity providers like Google, Facebook and Twitter, and more.

Chapter 6

Project Estimation and Schedule

6.1 System Estimation Plan

Sr No	Name	Description	Timeline	Remark
1	Requirement Analysis.	Complete specification of the system	19-Mar to 25-Mar	A detailed document should be there for each requirement.
2.	High-Level Modelling	Identifies the modules and different entities and their relationships	30-April to 28- May	Should decide on different modules and how they interact.
3.	Detailed Designing	GUI design, Program Specification		
4.	Construction	Code forsystem , Implementation of different modules	30-April to 28- May	Write code for different modules.
5.	Testing	Test the different modules together	04-June to 18-June	The system is tested using different test strategies
6.	Deployment		25-June	All requirements are fulfilled

Table 4: Project Estimation

6.2 Estimate

Project estimation and project scheduling are carried out together without accurate scheduling estimation there is no foundation for effective planning and support for rapid development. There are three parameters involved in computing the total cost of a software development project.

1. Software Cost.
2. Testing.
3. Efforts

Our development schedule consists of the following steps.

1. Estimation of the size of the product.
2. Estimation of the effort (Man- months).
3. Estimation of the schedule (Calendar month).

6.2.1 LOC based Estimation Effort.

Project Estimation report is size oriented matrix. Here constructive cost model i.e. COCOMO is used to estimate effort and time duration by using the size of the software.

The Constructive Cost Model (COCOMO) is an algorithmic software cost estimation model developed by Barry W. Boehm. This model uses a basic regression formula with parameters which are derived from the historical project data and current project characteristics. COCOMO consists of a hierarchy of the three increasingly detailed and accurate forms. The first level, Basic COCOMO is good for quick, early, rough order of magnitude estimates of software costs, but this model's accuracy is limited because of its lack of factors to account for the differences in project attributes (Cost Drivers). Intermediate COCOMO takes these Cost Drivers into the account and Detailed COCOMO having additionally accounts for the influence of individual project phases.

Sr. No:	File Name	Lines Of Code
1	App.vue	6
2	BookDetails.vue	254
3	BookTB.vue	14
4	Description.vue	16
5	History.vue	8
6	OrderSummary.vue	6
7	Error.vue	12
8	Checkbox.vue	25
9	Filter.vue	66
10	Header.vue	43
11	NavBarComponents.vue	34
12	Searchbar.vue	37
13	BookCoverCart.vue	30
14	LatsestBooks.vue	29

15	OrderCart.vue	149
16	OrderSummary.vue	30
17	ResultBook.vue	191
18	OrderDetails.vue	71
19	Siderbar.vue	63
20	OrderHistory.vue	83
21	Siderbar.vue	60
22	Profile.vue	99
23	Sidebar.vue	60
24	Profile.vue	142
25	Siderbar.vue	54
24	Wishbooks.vue	170
25	AddressPage.vue	159
26	LoginPage.vue	132
27	OtpPage.vue	159
28	RegisterPage.vue	179
29	cart.vue	263
30	Error.vue	28
31	BookView.vue	114
32	Home.vue	235
33	OrderDetails.vue	179
34	OrderHistory.vue	87
35	SearchResult.vue	229
36	UserProfileEdit.vue	40
37	UserProfileView.vue	69
38	WishList.vue	120
Total		3745

Table 5: LOC based Estimation Effort

```
dhond@Lightning MINGW64 /d/coding/e-commerce-website/src (main)
$ find . -name '*.vue' | xargs wc -l
     6 ./App.vue
   254 ./components/BookView/BookDetails.vue
    14 ./components/BookView/BookTB.vue
    16 ./components/BookView/Description.vue
     8 ./components/BookView/History.vue
     6 ./components/cart/OrderSummary.vue
    12 ./components/Error.vue
    25 ./components/Filter/Checkbox.vue
    66 ./components/Filter/Filter.vue
    43 ./components/Header/Header.vue
    34 ./components/Header/NavBarComponents.vue
    37 ./components/Header/SearchBar.vue
    30 ./components/HomePage/BookCoverCat.vue
    29 ./components/HomePage/LatestBooks.vue
   149 ./components/Search/OrderCart.vue
    30 ./components/Search/OrderSummary.vue
   191 ./components/Search/ResultBook.vue
    71 ./components/User/OrderDetailsView/OrderDetails.vue
    63 ./components/User/OrderDetailsView/Sidebar.vue
    83 ./components/User/OrderHistoryView/OrderHistory.vue
    60 ./components/User/OrderHistoryView/Sidebar.vue
    99 ./components/User/UserProfileEdit/Profile.vue
    60 ./components/User/UserProfileEdit/Sidebar.vue
   142 ./components/User/UserProfileView/Profile.vue
    54 ./components/User/UserProfileView/Sidebar.vue
   170 ./components/WishList/WishBooks.vue
   159 ./views/Auth/AddressPage.vue
   132 ./views/Auth/LoginPage.vue
   159 ./views/Auth/OtpPage.vue
   179 ./views/Auth/RegisterPage.vue
   263 ./views/Cart/cart.vue
    28 ./views/Error.vue
   114 ./views/HomePage/BookView.vue
   235 ./views/HomePage/home.vue
   179 ./views/OrderDetails/OrderDetails.vue
    87 ./views/OrderHistory/OrderHistory.vue
   229 ./views/SearchPage/SearchResult.vue
    40 ./views/UserProfile/UserProfileEdit.vue
    69 ./views/UserProfile/UserProfileView.vue
   120 ./views/WishList/WishList.vue
  3745 total
```

Fig 6.2.2:Lines of code per line with file location

6.3 Schedule

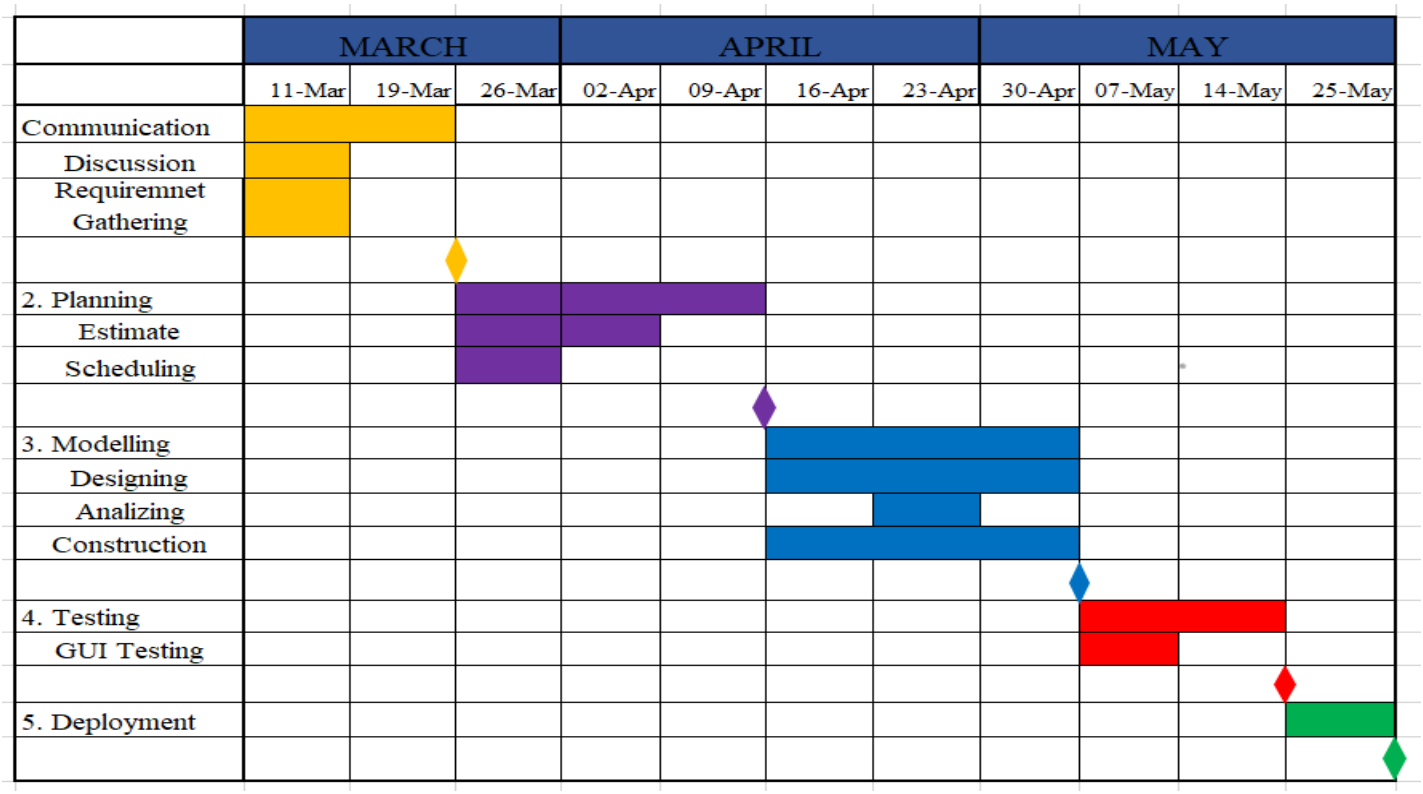

Fig. 6.3: Time Estimation Gantt Chart

Chapter 7

Software Testing

7.1 Introduction

Software Testing is that method of execution of system with the decided of finding errors. Software testing is an investigation aimed at evaluating certain attribute and capability of a program or system and predict that it meets its required specifications or not. Testing is to basically one of the means to detect software failures so that defects can be discovered and corrected as early as possible. Testing ensure that a all functions in software are properly working very well under all conditions and it establish that perform doesn't perform properly under specific conditions. Software system to wide it typically includes exhaustive checking of code along with execution of that code under various environments and conditions as well as examine the various aspects of code. In the today's culture of software development, a testing process is organized separately and parallel with the development process.

Software testing affects quality of software package directly as a result of derived from software testing may be used to correct the process by which software is developed. Software testing is a trade-off between budget, time and quality. In simple software Testing is defined as questioning a certain product in order to evaluate it.

7.2 Test Cases

7.2.1 Definition

A test case may be of conditions or variables under which a tester will determine whether a system under test satisfies requirements or works correctly. Testing is the process of running a system with the intention of finding errors. Testing enhances the integrity of a system by detecting deviations in design and errors in the system. Testing aims at detecting error-prone areas. This helps in the prevention of errors in a system. Testing also adds value to the product by conforming to the user requirements. The

main purpose of testing is to detect errors and error-prone areas in a system. Testing must be thorough and well-planned. A partially tested system is as bad as an untested system. And the price of an untested and under-tested system is high. The implementation is the final and important phase. It involves user-training, system testing in order to ensure successful running of the proposed system. The user tests the system and changes are made according to their needs. The testing involves the testing of the developed system using various kinds of data. While testing, errors are noted and correctness is the mode.

7.2.2 Fields in a Testcase

- **Test cast ID**
- **Unit to Test**
- **Prerequisite**
- **Test Data**
- **Test Procedure**
- **Expected Result**
- **Actual Results**
- **Status**
- **Remarks**

7.2.3 Testcase

Login

Sl No	Input Values	Test Case	Condition being checked	Result
1	Phone Number	Empty	Enter a valid Phone Number	Successful
2	Phone Number	Not in 10 digits	Enter a valid Phone Number	Successful
3	Full Name	Empty	Enter a valid Full Name	Successful
4	Full Name	Only First Name	Enter your Full Name	Successful
5	Email	Empty	Enter a valid email	Successful
6	Email	Not a valid email	Enter a valid email	Successful
7	Address	Empty	Address line 1, city/town/district, state, pin code can't be empty	Successful
8	OTP	Empty	Enter a valid OTP	Successful
9	OTP	Wrong OTP	Enter correct OTP number	Successful

Table 6: login Test Case Table

Search Bar

Sl No	Input Values	Test Case	Condition being checked	Result
1	Search Value	Empty	Enlist Random books	Successful
2	Search Values	Ravaan	Show book with name Raavan in it	Successful

Table 7: .Search Bar Test Case Table

Edit User Details

Sl No	Input Values	Test Case	Condition being checked	Result
1	Name	Empty	Enter a valid Full Name	Successful
2	Name	Only First Name	Enter your Full Name	Successful
3	Email	Empty	Enter a valid email	Successful
4	Email	Not a valid email	Enter a valid email	Successful
5	Address	Empty	Address line 1, city/town/district, state, pin code can't be empty	Successful

Table 8:Edit User Details Test Case Table

7.3 Unit Testing

Testing is performed by the developers before the system set-up is Transfer to the testing team to execute the test cases by using stubs and driver's. Unit testing is done by the respective developers to whom a module is assigned. The developers use test data that is separate from the test data of the quality assurance team. The purpose of unit testing is to separate each part of the program and ensure that individual parts are working correctly in terms of requirements and functionality. Performing unit testing on double-guard system following modules are tested individually.

- Pre-processing.
- Clustering.
- Post Clustering.
- Labelling Cluster.

Unit Testing Limitations

Testing cannot capture each error in an application. It is not possible to evaluate and test every execution path in every software application.

7.4 White Box Testing

In White box testing the detailed structure of internal logic, structure of the code is implemented. White box testing is also called glass box testing or open box testing. White box testing is application the tester necessary to have good domain knowledge as well as tester should be aware about the internal working of the code, loops. The tester necessary to have a see inside the source code and search out which unit of the code is behaving inappropriately.

7.5 Black Box Testing

Testing method while not having any data of internal working of the applying is Black Box testing. The tester is completely new one to the system architecture. Black box tests a tester will interact with the systems user interface provides inputs and examining outputs while not knowing however and where the inputs are worked upon.

Chapter 8

Future Scope

- **Conventional Buy**
  Without moving out from home people can easily buy the books as per their wish.

- **Delivery**
  User will get ordered books as per their convenient address

- **More Outreach**
  People across the country can buy books easily.

- **Increase the Reach Of Yogabhyas Mandal through books**
  As their primary aim is to obtain contentment, happiness as well as a healthy body, mind and soul.

- **Keep Interest in Books**
  Ebook store will help people to be in touch with books and let them to know about the incoming books.

- **Growth of New Authors**
  People who are new writers get a platform to publish their books which will help them to grow.

- **Crazy of ebook store**.
  Due to digital technologies people will more move towards the ebook store to buy books.

- **Web Availability For Books As it is open to all.**

Chapter 9

Project Output

- **Login Page**

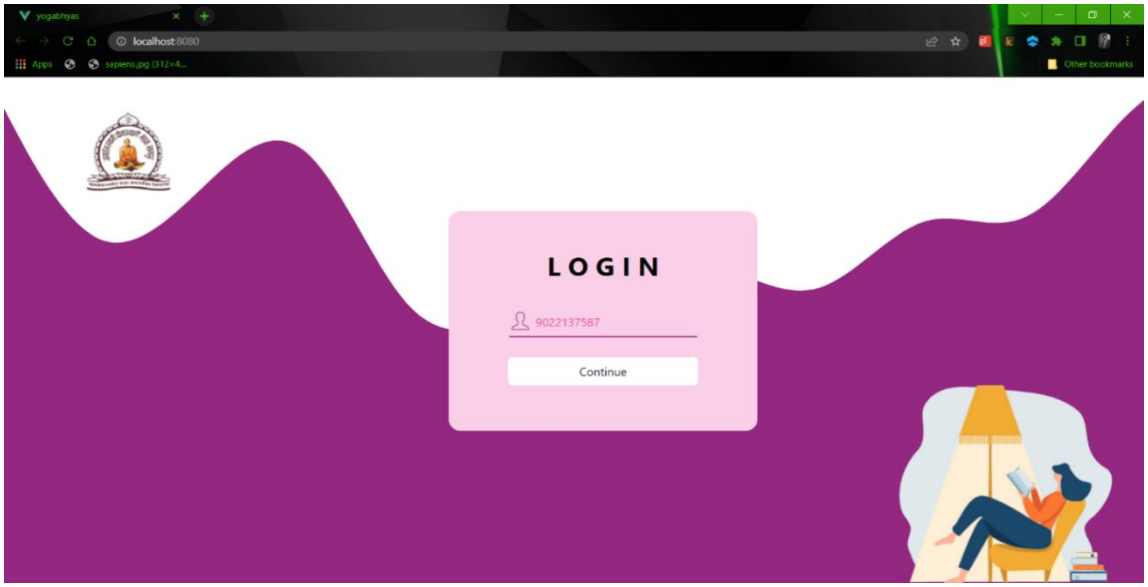

Fig 9.1 Login Page

- **Register Page**

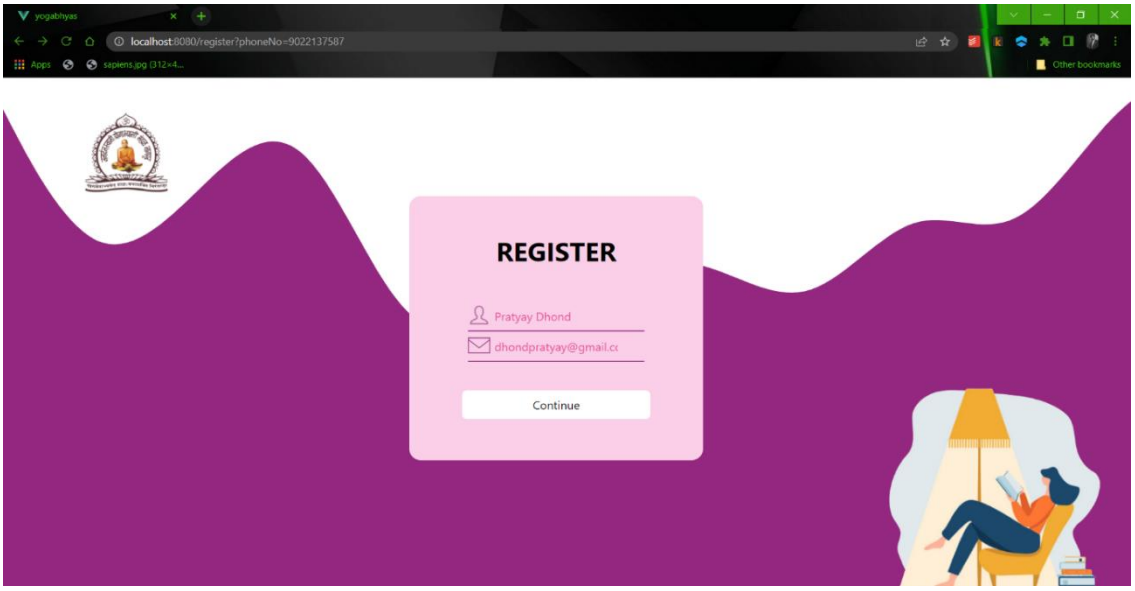

Fig 9.2 Register Page

- **Address Page**

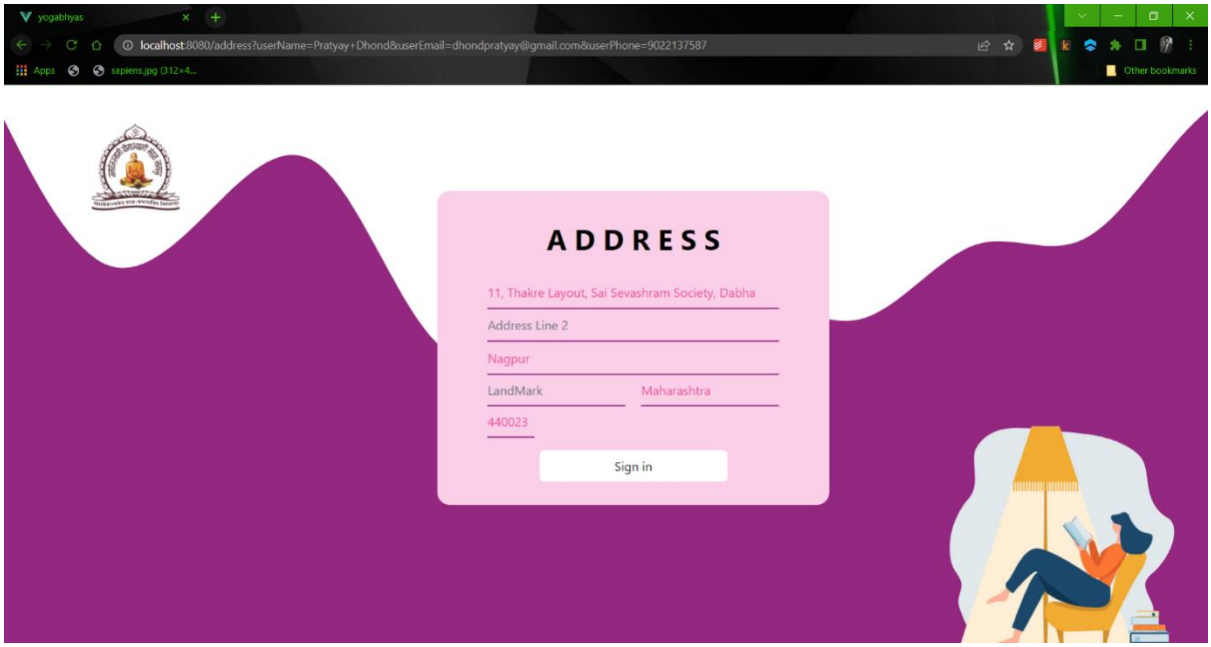

Fig 9.3 Address Page

- **OTP Page**

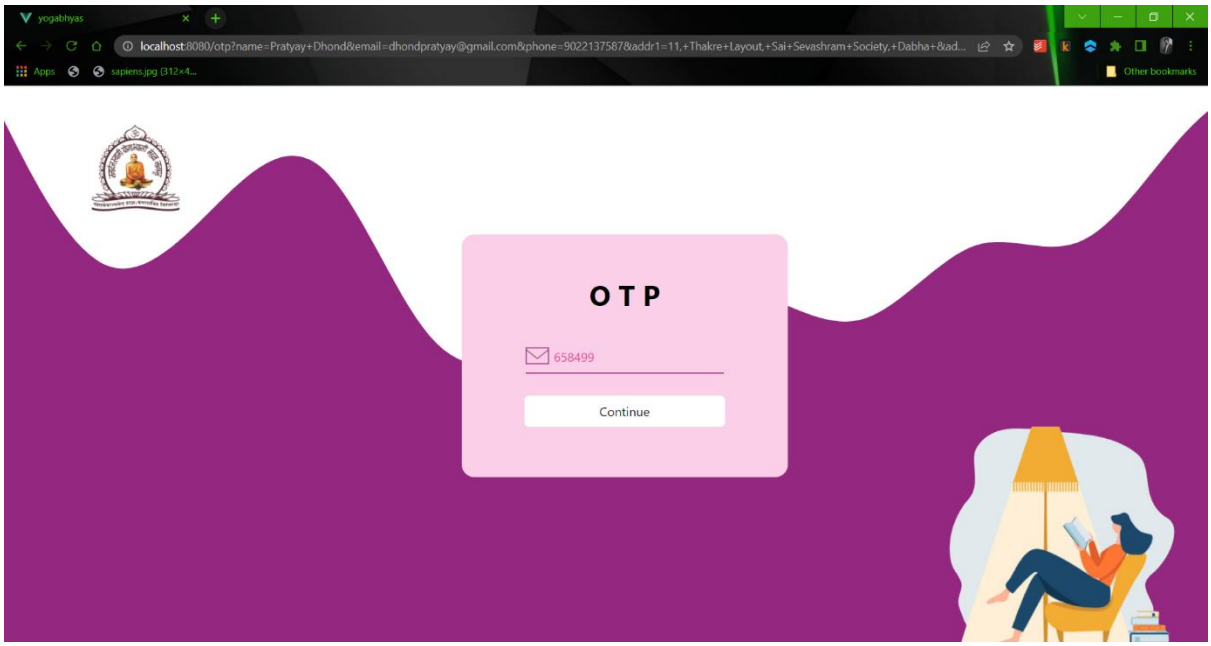

Fig 9.4 OTP Page

- **Home Page**

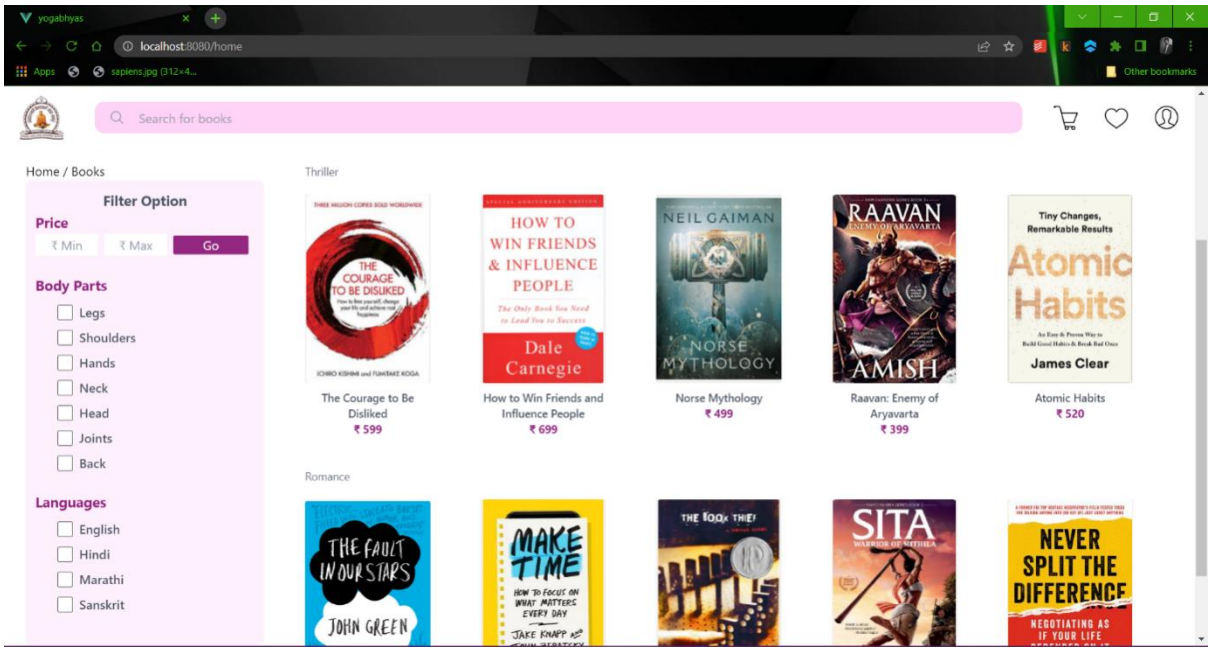

Fig 9.5 Home Page

- **Book Page**

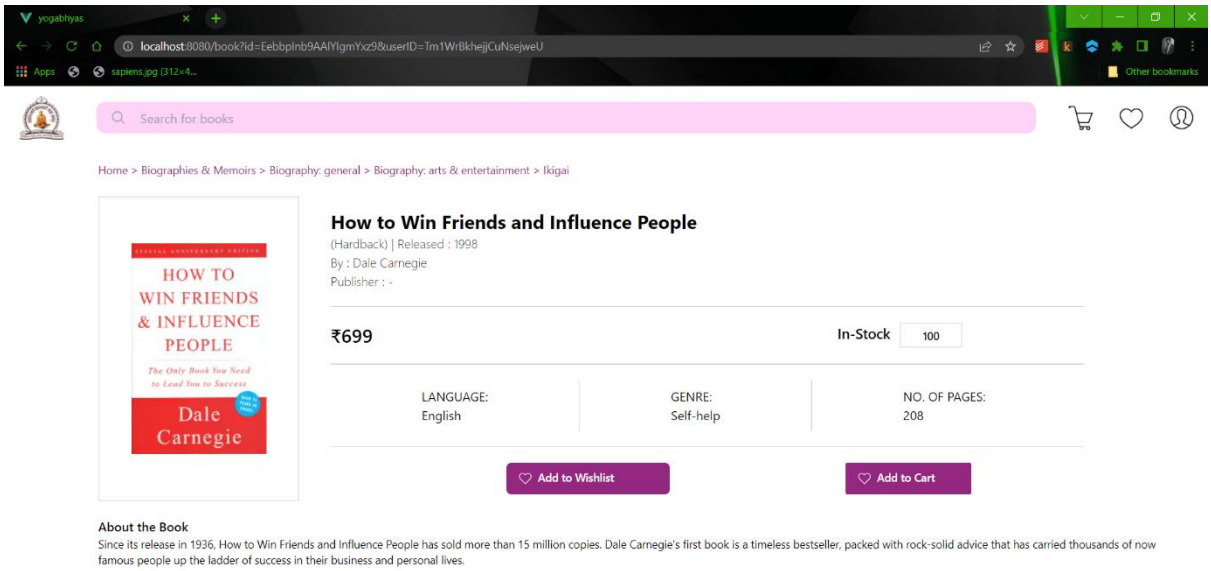

Fig 9.6 Book Page

- **Search Result Page**

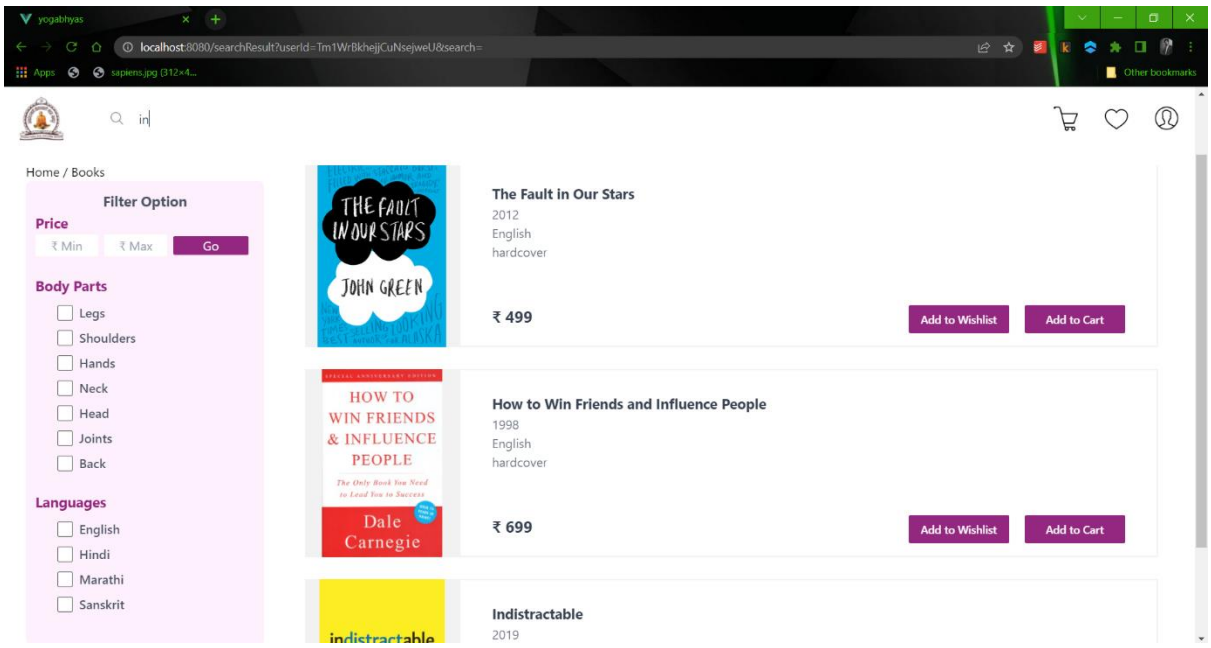

Fig 9.7 Search Result Page

- **Cart Page**

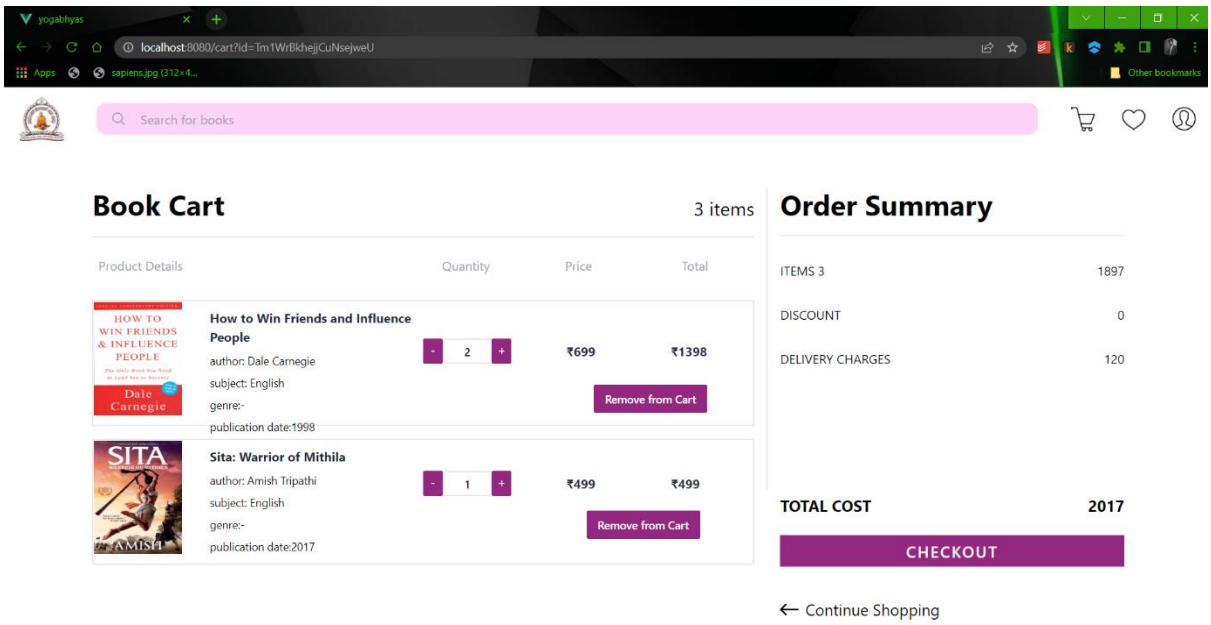

Fig 9.8 Cart Page

- **Wishlist Page**

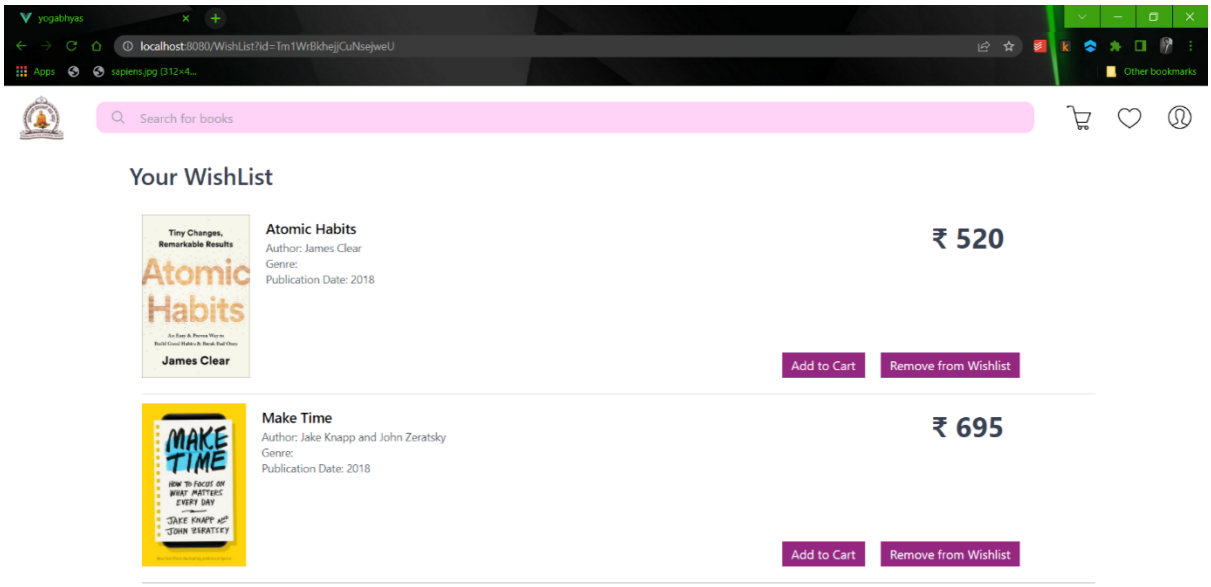

Fig 9.9 WishList Page

- **User Profile Page**

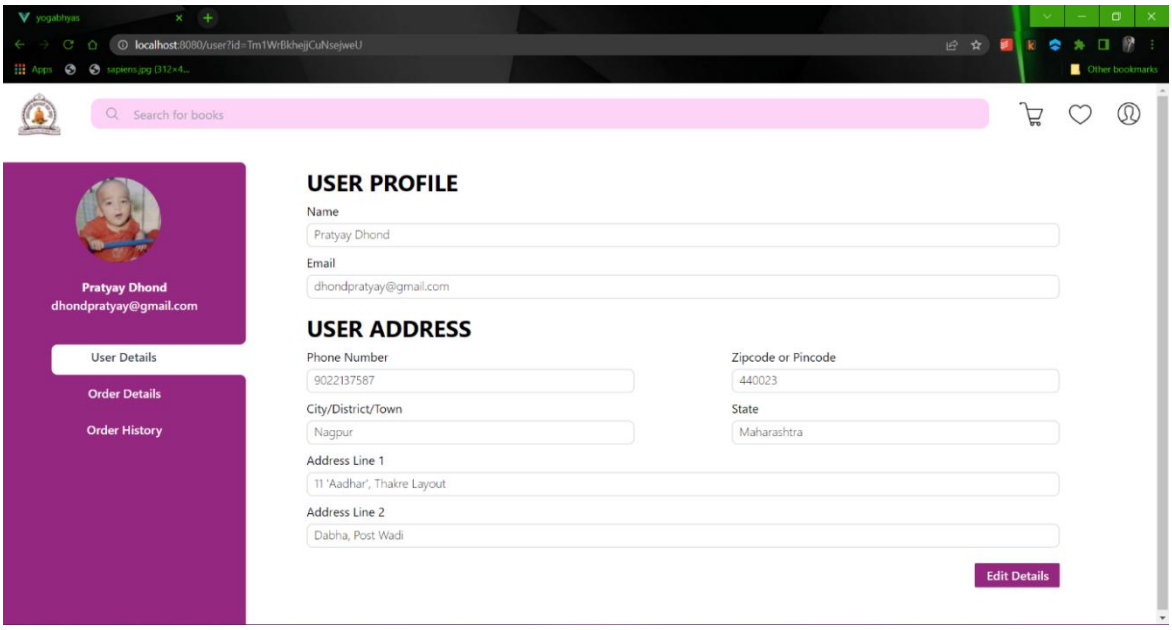

Fig 9.10 User Profile Page

- **User Profile Edit Page**

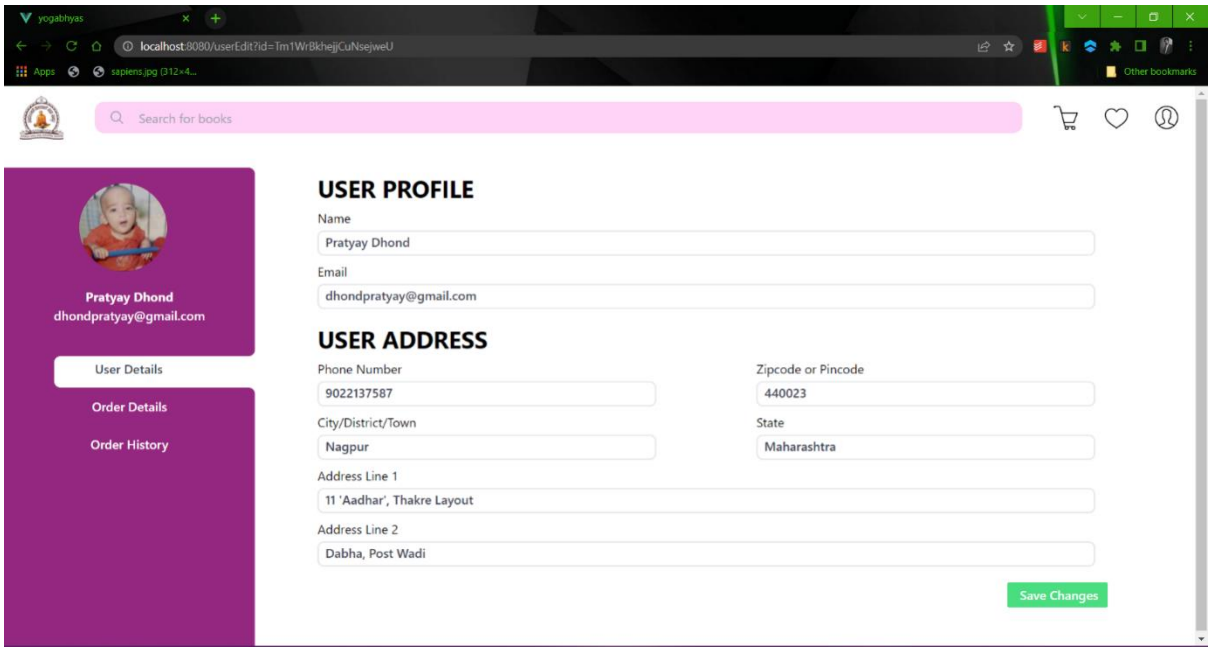

Fig 9.11 User Profile Edit Page

Chapter 10

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  - https://www.amazon.in/?tag=msndeskabkin-21&hvadid=72705283629710&hvqmt=e&hvbmt=be&hvdev=c&ref=pd_sl_7qhce485bd_e

- **YouTube**
  - https://www.youtube.com/watch?v=YrxBCBibVo0&list=PL4cUxeGkcC9hYYGbV60Vq3IXYNfDk8At1
  - https://www.youtube.com/watch?v=FMPHvxqDrVk&t=382s